

Homework 1

Problem 1. Rossby Wave Instability (50 points)

Consider a two-dimensional, inviscid, non-magnetic disk in cylindrical coordinates (r, ϕ) . Neglect disk self-gravity. Let Σ denote surface density and P vertically integrated pressure. Adopt the barotropic equation of state $P = c_s^2 \Sigma$, with constant c_s . The governing equations are

$$\frac{\partial \Sigma}{\partial t} + \nabla \cdot (\Sigma \mathbf{v}) = 0, \quad (1)$$

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{\nabla P}{\Sigma} - \nabla \Phi, \quad \Phi = -\frac{GM_*}{r}. \quad (2)$$

The axisymmetric equilibrium has $v_{r0} = 0$ and $v_{\phi0} = r\Omega(r)$, with

$$\Sigma_0(r) = \Sigma_b \left[1 + A \exp \left(-\frac{(r - r_0)^2}{2w^2} \right) \right], \quad A > 0, \quad w \ll r_0. \quad (3)$$

The rotation profile must satisfy radial force balance; it is *not* to be prescribed independently as exactly Keplerian. Define

$$\kappa^2 = \frac{1}{r^3} \frac{d(r^4 \Omega^2)}{dr}, \quad \zeta_0 = \frac{\kappa^2}{2\Omega \Sigma_0}. \quad (4)$$

Consider non-axisymmetric perturbations proportional to $e^{i(m\phi - \omega t)}$, with integer $m \geq 1$. Define the Doppler-shifted frequency $\tilde{\omega}(r) = \omega - m\Omega(r)$ and the enthalpy perturbation $W(r) = \delta P / \Sigma_0$.

1. **Equilibrium and linearization (15 points).** Derive $\Omega^2(r)$ from radial force balance and write the linearized continuity, radial-momentum, and azimuthal-momentum equations. State the conditions needed for $\Omega^2 > 0$ and $\kappa^2 > 0$.
2. **Radial eigenvalue equation (20 points).** Eliminate δv_r , δv_ϕ , and $\delta \Sigma$ to obtain a second-order ordinary differential equation for $W(r)$. Write every coefficient explicitly in terms of background quantities and ω . Show where the vortensity gradient enters the equation and identify the corotation radius r_c , defined by $\text{Re}(\omega) = m\Omega(r_c)$.
3. **Boundary conditions and instability (10 points).** Consider a finite domain $r_{\text{in}} < r < r_{\text{out}}$ enclosing the bump, with impermeable boundaries $\delta v_r = 0$. Translate these conditions into conditions on W . Explain why ω is an eigenvalue and give the growth condition under the stated time convention. Distinguish the role of a vortensity extremum from a sufficient condition for an unstable global mode.
4. **Physical interpretation (5 points).** Explain how Rossby waves on opposite sides of a vortensity extremum can interact. Explain why a density bump alone does not justify assuming that the disk is unstable, and distinguish RWI from axisymmetric centrifugal instability.

Problem 2. Magnetized VSI in Ideal MHD (50 points)

Use a local Cartesian frame (x, y, z) rotating at angular frequency Ω , with x , y , and z pointing in the radial, azimuthal, and vertical directions. Adopt an inviscid, incompressible local model with uniform reference density ρ_0 and ideal MHD. Neglect effective buoyancy, as appropriate to the rapid thermal-relaxation limit; no energy equation or non-ideal magnetic terms are required. Background force balance is assumed to arise from a baroclinic disk; retain its local shear coefficients in the perturbation equations.

Take the background flow and uniform poloidal magnetic field to be

$$\mathbf{U}_0 = (-q\Omega x + Sz)\mathbf{e}_y, \quad \mathbf{B}_0 = B_x\mathbf{e}_x + B_z\mathbf{e}_z, \quad B_z \neq 0, \quad (5)$$

where $q = -d \ln \Omega / d \ln R$, $0 < q < 2$, and $S = \partial_z U_{0y}$ is the local vertical shear. Treat Ω , q , and S as constants. The reference point is away from the disk midplane when $S \neq 0$.

Consider axisymmetric perturbations proportional to $e^{st+i(k_x x + k_z z)}$, with $k_z \neq 0$. Use the definitions

$$\mathbf{u} = \delta \mathbf{v}, \quad \mathbf{b} = \frac{\delta \mathbf{B}}{\sqrt{4\pi\rho_0}}, \quad \Pi = \frac{\delta P}{\rho_0} + \frac{\mathbf{B}_0 \cdot \delta \mathbf{B}}{4\pi\rho_0}, \quad (6)$$

$$\mu = \frac{k_x}{k_z}, \quad \omega_A = \frac{\mathbf{k} \cdot \mathbf{B}_0}{\sqrt{4\pi\rho_0}}, \quad \kappa^2 = 2(2 - q)\Omega^2. \quad (7)$$

Here Π is the perturbed total pressure per unit mass. *The background magnetic-field components cannot be chosen independently if the background is stationary.*

1. **Consistency of the background (10 points).** Starting from

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{v} \times \mathbf{B}), \quad \nabla \cdot \mathbf{B} = 0,$$

derive the condition on B_x/B_z for a time-independent background. Explain why a purely vertical field is not a stationary background when $S \neq 0$. Use a background satisfying the condition you derive for the remaining parts.

2. **Linearized equations (15 points).** Derive the three velocity and three induction equations, retaining Coriolis forces, radial shear, vertical shear, and magnetic tension. Use $\mathbf{k} \cdot \mathbf{u} = \mathbf{k} \cdot \mathbf{b} = 0$ to eliminate u_z and b_z . Eliminate Π and write the resulting four-variable system for (u_x, u_y, b_x, b_y) .

3. **Dispersion relation (15 points).** Set the determinant of this system to zero and obtain a polynomial in s . Write its coefficients explicitly in terms of Ω , κ , S , μ , and ω_A . Account for the background-field constraint when interpreting ω_A . Verify the hydrodynamic VSI limit $\omega_A \rightarrow 0$ and the classical vertical-field MRI limit $S = 0$, $k_x = 0$.

4. **Growth and interpretation (10 points).** Derive the hydrodynamic VSI growth condition. For the magnetized problem, determine the condition for a real positive root s^2 and discuss whether magnetic tension always stabilizes the flow. Distinguish a branch connected continuously to hydrodynamic VSI from MRI growth driven by radial shear; do not label every growing magnetic mode as VSI.

Suggested reading.

Lovelace et al. (1999), [arXiv:astro-ph/9809321](#).

Li et al. (2000), [arXiv:astro-ph/9907279](#).

Balbus & Hawley (1991), ApJ, 376, 214.

Lin & Youdin (2015), [arXiv:1505.02163](#).

The local magnetized model in Problem 2 is a deliberately simplified ideal-MHD problem, rather than a full vertically global disk model.